Crossing Symmetry: Compton → Pair Annihilation

Griffiths Convention (s- and t-channel Compton)

Griffiths 7.30/7.41/7.47 | H&M §6.4 | Thomson §6.5.6

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1 Convention Alert: Griffiths vs. Other Texts

The Labeling Issue

Different textbooks number the Compton scattering particles differently, which changes whether the second diagram is called "t-channel" or "u-channel." **The physics is identical**—it is the same propagator, same momentum—only the *name* changes.

Convention	**1**	**2**	**3**	**4**
Griffiths	e^-_{in}	γ_{out}	γ_{in}	e^-_{out}
Standard (P&S, L&B)	e^-_{in}	γ_{in}	e^-_{out}	γ_{out}

Consequence:

- **Griffiths:** Second propagator is $(p_1 - p_2)^2$. With $t = (p_1 - p_2)^2$, this is the **t-channel**.
  $\Rightarrow$ Compton diagrams are **s-channel and t-channel**.
- **Standard (P&S, L&B):** Same propagator is $(p_1 - p_4)^2 = u$, so it is called **u-channel**.
  $\Rightarrow$ Compton diagrams are **s-channel and u-channel**.

We use Griffiths' convention throughout this document.

2 Compton Scattering in Griffiths' Labeling

$$\boxed{\gamma(p_3) + e^-(p_1) \longrightarrow \gamma(p_2) + e^-(p_4)}$$

Griffiths' Particle Assignments (Example 7.4)

Particle 1: e^- incoming, momentum p_1, spinor $u(1)$
Particle 2: γ outgoing, momentum p_2, polarization ε_2
Particle 3: γ incoming, momentum p_3, polarization ε_3
Particle 4: e^- outgoing, momentum p_4, spinor $\bar{u}(4)$

Mandelstam variables (Griffiths' convention):

$$s = (p_1 + p_3)^2 = (p_2 + p_4)^2 \qquad \text{(sum of incoming momenta)} \tag{1}$$

$$t = (p_1 - p_2)^2 = (p_3 - p_4)^2 \qquad \text{(incoming } e^- \text{ to outgoing } \gamma\text{)} \tag{2}$$

$$u = (p_1 - p_4)^2 = (p_3 - p_2)^2 \qquad \text{(incoming } e^- \text{ to outgoing } e^-\text{)} \tag{3}$$

Constraint: $s + t + u = 2m^2$ (two photons massless).

2.1 The Two Diagrams (Griffiths Eqs. 7.111–7.112)

Diagram 1: s-channel

e^- absorbs γ_{in}, propagates, emits γ_{out}

Internal momentum: $p_1 + p_3$

Propagator: $\dfrac{1}{(p_1+p_3)^2 - m^2} = \dfrac{1}{s - m^2}$

Griffiths Eq. 7.111:

$$\mathcal{M}_s \propto \frac{\bar{u}(4)\,\not{\epsilon}_2\,(\not{p}_1 + \not{p}_3 + m)\,\not{\epsilon}_3\,u(1)}{s - m^2}$$

Diagram 2: t-channel

e^- emits γ_{out}, propagates, absorbs γ_{in}

Internal momentum: $p_1 - p_2$

Propagator: $\dfrac{1}{(p_1-p_2)^2 - m^2} = \dfrac{1}{t - m^2}$

Griffiths Eq. 7.112:

$$\mathcal{M}_t \propto \frac{\bar{u}(4)\,\not{\epsilon}_3\,(\not{p}_1 - \not{p}_2 + m)\,\not{\epsilon}_2\,u(1)}{t - m^2}$$

Why Griffiths Gets s and t Channels

In Griffiths' labeling, the outgoing *photon* is particle 2. So $t = (p_1 - p_2)^2$ computes the squared momentum transfer from the incoming electron to the outgoing photon. The second diagram has exactly this momentum flowing through the propagator.
In the standard convention (P&S), the outgoing photon is particle 4, so the *same* momentum transfer would be $u = (p_1 - p_4)^2$. Same propagator, different name.

2.2 The Compton Result in Griffiths' Variables

Setting $m = 0$ and using the $-g_{\mu\nu}$ polarization sum (H&M Eq. 6.111):

Compton $\overline{|\mathcal{M}|^2}$ in Griffiths' Convention

$$\overline{|\mathcal{M}|^2}_{\text{Compton}} = -2e^4\left(\frac{t}{s} + \frac{s}{t}\right) = -2e^4\,\frac{s^2 + t^2}{st} \tag{4}$$

The two terms correspond to the two channels:

- s-channel contributes: $-2e^4\,(t/s)$
- t-channel contributes: $-2e^4\,(s/t)$
- Interference: 0 (at $m = 0$, by helicity conservation)

Translating Between Conventions

This is the *same physics* as the Lancaster & Blundell result $-2e^4(u/s + s/u)$. The mapping is:

$$t_{\text{Griffiths}} = u_{\text{L\&B}}, \qquad u_{\text{Griffiths}} = t_{\text{L\&B}}$$

since Griffiths swaps the labels of the two outgoing particles relative to the standard convention. The variable s is the same in both.

3 Crossing Symmetry: The Big Idea

Crossing Symmetry (Griffiths Section 1.4)

An **incoming particle** with 4-momentum p can be "crossed" to become an **outgoing antiparticle** with 4-momentum $-p$ (and vice versa). The matrix element is the *same analytic function* of the momenta.

Griffiths: "*The power and beauty of crossing symmetry can scarcely be exaggerated. It tells us that Compton scattering $\gamma + e^- \to \gamma + e^-$ is 'really' the same process as pair annihilation $e^+ + e^- \to \gamma + \gamma$, although in the laboratory they are completely different phenomena.*"

The rule: To cross a leg, negate its 4-momentum and flip particle $\leftrightarrow$ antiparticle. Since the photon is its own antiparticle, crossing a photon just negates its momentum.

4 The Crossing: Compton $\to$ Pair Annihilation

4.1 The Two Processes Side by Side

$$\textbf{Compton:} \quad \gamma(p_3) + e^-(p_1) \to \gamma(p_2) + e^-(p_4)$$

$$\textbf{Pair Annihilation:} \quad e^-(q_1) + e^+(q_2) \to \gamma(q_3) + \gamma(q_4)$$

(I use q_i for annihilation momenta to avoid confusion with the Compton p_i.)

4.2 Which Legs to Cross

To turn Compton into pair annihilation, we make two crossings:

The Two Crossings

Cross 1: Incoming $\gamma(p_3)$ $\to$ outgoing $\gamma(q_3)$

$$p_3 \longrightarrow -q_3$$

The incoming photon becomes an outgoing photon (photon is its own antiparticle). It moves from the left side to the right side of the reaction.

Cross 2: Outgoing $e^-(p_4)$ $\to$ incoming $e^+(q_2)$

$$p_4 \longrightarrow -q_2$$

The outgoing electron becomes an incoming positron (electron $\leftrightarrow$ positron under crossing).

Remaining legs (unchanged):

$$p_1 \longrightarrow q_1 \quad \text{(incoming } e^- \text{ stays incoming } e^-\text{)}$$

$$p_2 \longrightarrow q_4 \quad \text{(outgoing } \gamma \text{ stays outgoing } \gamma\text{)}$$

Complete Momentum Map

$$\boxed{p_1 \to q_1, \qquad p_2 \to q_4, \qquad p_3 \to -q_3, \qquad p_4 \to -q_2} \tag{5}$$

4.3 How the Diagrams Transform

	Compton	Annihilation
Diagram 1 propagator	$(p_1 + p_3)^2 = s_C$	$(q_1 - q_3)^2 = t_A$
Diagram 2 propagator	$(p_1 - p_2)^2 = t_C$	$(q_1 - q_4)^2 = u_A$

Diagram Topology Change

- The Compton s-**channel** (both incoming particles enter the propagator) becomes an annihilation t-**channel** (electron emits one photon, then the other). This is because the incoming photon p_3 got crossed to an outgoing photon q_3, so $p_1 + p_3$ became $q_1 - q_3$: a momentum *transfer*, not a total energy.
- The Compton t-**channel** becomes an annihilation u-**channel**: $p_1 - p_2$ became $q_1 - q_4$ (electron emits the *other* photon first).
- There is **no** s-**channel** in pair annihilation at tree level! You cannot have $e^+e^- \to$ single virtual electron $\to \gamma\gamma$, because the QED vertex connects two fermion lines and one photon line—you can't have a single fermion absorb two photons at one vertex.

5 What Happens to the Mandelstam Variables

This is the punchline. Apply the momentum map (5) to each Compton Mandelstam variable:

5.1 Detailed Calculation

Compton s:

$$s_C = (p_1 + p_3)^2 \xrightarrow{p_3 \to -q_3} (q_1 - q_3)^2 = t_A \tag{6}$$

Compton t:

$$t_C = (p_1 - p_2)^2 \xrightarrow{p_2 \to q_4} (q_1 - q_4)^2 = u_A \tag{7}$$

Compton u:

$$u_C = (p_1 - p_4)^2 \xrightarrow{p_4 \to -q_2} (q_1 + q_2)^2 = s_A \tag{8}$$

Mandelstam Variable Mapping (Griffiths Convention)

$$\boxed{s_C \longrightarrow t_A, \qquad t_C \longrightarrow u_A, \qquad u_C \longrightarrow s_A} \tag{9}$$

In words: $s \to t$, $t \to u$, $u \to s$ — a **cyclic permutation**.

Why $s \to t$ Makes Physical Sense

In Compton, $s_C = (p_1 + p_3)^2$ is the squared total energy of the incoming $e^-\gamma$ pair. After crossing the photon to the final state, this becomes $t_A = (q_1 - q_3)^2$, a squared momentum

transfer—because what was an incoming particle is now outgoing.
Similarly, $u_C = (p_1 - p_4)^2$ (incoming e^- vs. outgoing e^-) becomes $s_A = (q_1 + q_2)^2$ (total CM energy of the e^+e^- pair) because the outgoing e^- got crossed to an incoming e^+.

6 Applying the Crossing to $\overline{|\mathcal{M}|^2}$

6.1 Starting Point: Compton (Griffiths Convention)

$$\overline{|\mathcal{M}|^2}_{\text{Compton}} = -2e^4\left(\frac{t_C}{s_C} + \frac{s_C}{t_C}\right) \tag{10}$$

6.2 Apply the Map $s_C \to t_A$, $t_C \to u_A$

$$\overline{|\mathcal{M}|^2}_{\text{Compton}}\Big|_{\substack{s_C\to t_A \\ t_C\to u_A}} = -2e^4\left(\frac{u_A}{t_A} + \frac{t_A}{u_A}\right) \tag{11}$$

6.3 Averaging Factor Check

- **Compton:** $\frac{1}{4}$ (average over 2 electron spins $\times$ 2 photon polarizations).
- **Pair annihilation:** $\frac{1}{4}$ (average over 2 electron spins $\times$ 2 positron spins).

Both give $\frac{1}{4}$, so no correction is needed.

Pair Annihilation Result via Crossing (Griffiths Problem 7.41, $m = 0$)

$$\boxed{\overline{|\mathcal{M}|^2}_{e^+e^-\to\gamma\gamma} = -2e^4\left(\frac{u}{t} + \frac{t}{u}\right) = -2e^4\,\frac{t^2+u^2}{tu}} \tag{12}$$

where t and u now refer to the annihilation Mandelstam variables (subscript A dropped).

Physical region: $s_A > 0,\ t_A \le 0,\ u_A \le 0$. So $-u/t > 0$ and $-t/u > 0 \Rightarrow \overline{|\mathcal{M}|^2} > 0$. ✓

Note: the annihilation has t- **and** u-**channel** propagators, matching the fact that t and u (not s) appear in the denominators.

7 Pair Annihilation Amplitudes: Direct Verification

Pair Annihilation Amplitudes (Griffiths Eqs. 7.133–7.134)

For $e^-(q_1) + e^+(q_2) \to \gamma(q_3) + \gamma(q_4)$:
t-channel:

$$\mathcal{M}_t \propto \frac{\bar{v}(q_2)\,\not{\epsilon}_4\,(\not{q}_1 - \not{q}_3 + m)\,\not{\epsilon}_3^*\,u(q_1)}{(q_1 - q_3)^2 - m^2} = \frac{\cdots}{t - m^2}$$

u-channel (photons swapped):

$$\mathcal{M}_u \propto \frac{\bar{v}(q_2)\,\not{\epsilon}_3^*\,(\not{q}_1 - \not{q}_4 + m)\,\not{\epsilon}_4\,u(q_1)}{(q_1 - q_4)^2 - m^2} = \frac{\cdots}{u - m^2}$$

Comparing with Compton under crossing:
The Compton s-channel amplitude (Griffiths 7.111) is:

$$\mathcal{M}_s^C \propto \frac{\bar{u}(p_4)\,\not{\varepsilon}_2\,(\not{p}_1 + \not{p}_3 + m)\,\not{\varepsilon}_3\,u(p_1)}{(p_1+p_3)^2 - m^2}$$

Under the crossing $p_1 \to q_1$, $p_3 \to -q_3$, $p_4 \to -q_2$, $p_2 \to q_4$:

- $\bar{u}(p_4) \to \bar{u}(-q_2) \sim \bar{v}(q_2)$ (outgoing particle $\to$ incoming antiparticle)
- $\not{p}_1 + \not{p}_3 \to \not{q}_1 - \not{q}_3$ (propagator momentum)
- $(p_1+p_3)^2 - m^2 \to (q_1 - q_3)^2 - m^2 = t_A - m^2$
- $\varepsilon_3 \to \varepsilon_3^*$ (incoming $\to$ outgoing photon conjugates polarization)

This exactly reproduces the pair annihilation t-channel amplitude. ✓

The $\bar{u} \to \bar{v}$ Magic

The key subtlety: when the outgoing $e^-(p_4)$ is crossed to incoming $e^+(q_2)$, the spinor changes from $u(p_4)$ to $v(q_2)$ with $p_4 = -q_2$. In the completeness relations, $\sum u\bar{u} = \not{p} + m$ while $\sum v\bar{v} = \not{p} - m$. But since $p_4 = -q_2$:

$$\not{p}_4 + m = -\not{q}_2 + m = -(\not{q}_2 - m)$$

The sign of m flips automatically when you negate the momentum. This is why crossing "just works" at the level of $|\mathcal{M}|^2$ expressed in Mandelstam variables: the Mandelstam substitution encodes *both* the momentum change and the particle/antiparticle swap.

8 Invariant Cross Section $d\sigma/dt$

Using $d\sigma/dt = \overline{|\mathcal{M}|^2}/(16\pi s^2)$ for massless particles:

Invariant Cross Sections

Compton (Griffiths convention):

$$\left.\frac{d\sigma}{dt}\right|_{\text{Compton}} = \frac{-2e^4}{16\pi s^2}\left(\frac{t}{s} + \frac{s}{t}\right) = -\frac{2\pi\alpha^2(s^2+t^2)}{s^3\,t} \tag{13}$$

Pair Annihilation:

$$\boxed{\left.\frac{d\sigma}{dt}\right|_{e^+e^-\to\gamma\gamma} = -\frac{2\pi\alpha^2(t^2+u^2)}{s^2\,tu}} \tag{14}$$

9 The Full Crossing Web

All Three Related Processes

Process	Propagator channels	$\overline{\lvert\mathcal{M}\rvert^2}$	From Compton by
$\gamma e^- \to \gamma e^-$ (Compton)	$s,\ t$	$-2e^4\left(\frac{t}{s}+\frac{s}{t}\right)$	— (direct)
$e^+e^- \to \gamma\gamma$ (Annihilation)	$t,\ u$	$-2e^4\left(\frac{u}{t}+\frac{t}{u}\right)$	$s\to t,\ t\to u,\ u\to s$
$\gamma\gamma \to e^+e^-$ (Pair production)	$t,\ u$	$-2e^4\left(\frac{u}{t}+\frac{t}{u}\right)$	time-reverse of ann.

Pattern: The result always involves exactly the two Mandelstam variables labeling the propagator channels. Compton has s- and t-channel propagators $\Rightarrow$ s and t appear. Annihilation has t- and u-channels $\Rightarrow$ t and u appear.

10 Summary: The Crossing Recipe

Step-by-Step Crossing Recipe (Griffiths Convention)

1. **Label.** Use Griffiths' labeling for Compton: $1 = e^-_{\text{in}}$, $2 = \gamma_{\text{out}}$, $3 = \gamma_{\text{in}}$, $4 = e^-_{\text{out}}$.
2. **Identify crossings.** To get pair annihilation $e^-(q_1)e^+(q_2) \to \gamma(q_3)\gamma(q_4)$:
$$p_1 \to q_1, \quad p_2 \to q_4, \quad p_3 \to -q_3, \quad p_4 \to -q_2$$
3. **Map Mandelstam variables.** Substitute into s_C, t_C, u_C:
$$s_C \to t_A, \quad t_C \to u_A, \quad u_C \to s_A \qquad (\text{cyclic: } s \to t \to u \to s)$$
4. **Substitute into $\overline{\lvert\mathcal{M}\rvert^2}$.**
$$-2e^4\left(\frac{t}{s}+\frac{s}{t}\right) \xrightarrow{s\to t,\ t\to u} -2e^4\left(\frac{u}{t}+\frac{t}{u}\right)$$
5. **Verify:** averaging factor ($\frac{1}{4}$ in both cases), positivity in the physical region.
6. **Cross section:** $\dfrac{d\sigma}{dt} = \dfrac{\overline{\lvert\mathcal{M}\rvert^2}}{16\pi\, s^2}$.

The Deep Lesson

Crossing symmetry means the S-matrix is a single analytic function of the Mandelstam variables. Evaluating it in different kinematic regions ($s > 0$ vs. $t > 0$ vs. $u > 0$) describes physically distinct processes. Compton scattering and pair annihilation look completely different in the

lab, but they are the *same amplitude* evaluated at different kinematic points.
The channel labels (s, t, or u) are just names for which Mandelstam variable is timelike (> 0) in each process:

- **Compton:** $s > 0$ (CM energy of $e^-\gamma$ system)
- **Annihilation:** $s > 0$ (CM energy of e^+e^- system), but now s_A corresponds to what was u_C